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$$\begin{aligned} & \int \frac{x^2 - 2x + 4}{x^2 - 2x + 1} dx \\ &= \int \frac{(x-1)^2 + 3}{(x-1)^2} dx \\ &= \int \frac{(x-1)^2}{(x-1)^2} dx + \int \frac{3}{(x-1)^2} dx \\ &= \int 1 dx + 3 \int \frac{1}{(x-1)^2} dx \\ & t = x - 1 \Rightarrow dt = dx \\ &= \int 1 dx + 3 \int \frac{1}{t^2} dt \\ &= \int 1 dx + 3 \int t^{-2} dt \\ &= x - 3t^{-1} + C \\ &= x - \frac{3}{x-1} + C \end{aligned}$$

References